

Enterprise Use Case Variants — All Assignments

Domain Data · Instruction Pairs · Models · Guidance for 1A, 1B, 2A, 2B & 2C

How to use this guide:

1. Pick any variant (V1–V6). Your domain data sources, instruction pairs listed under that variant.
2. **Students can choose their respective enterprise use case. In that case, label it as a custom enterprise variant. You are required to create the template for your custom use case in the same format as given for V2 -V6 and submit as a separate doc.**
3. Pipeline steps, marking rubrics, and submission requirements are fixed per assignment and do NOT change across variants. Only domain data, instruction pairs, models, and technique choices differ.
4. EL-2 assignments (2A, 2B, 2C) all reuse the same .txt files and instruction_dataset.jsonl you collected for your EL-1 assignment. No new data collection is required.
5. **Students are encouraged to experiment with different models other than recommended**

Variant 1 — Default — Domain Q&A Assistant

Open Choice · Medical · Legal · Financial · HR · Manufacturing · Scientific

This is the base variant for all assignments. Select any domain from the model selection table in your assignment document, collect domain PDFs, and build a domain Q&A assistant. No additional domain-specific guidance is needed — refer directly to each assignment document for pipeline steps and marking criteria.

Variant 2 — Troubleshooting Assistant for Customer Support

Telecom / SaaS / Cloud Infrastructure — Support Ticket Deflection

A telecom or SaaS company receives thousands of Tier-1 support tickets daily for known, documented issues. The enterprise wants an LLM-powered assistant that reads a user's symptom description and generates accurate, step-by-step troubleshooting instructions — reducing agent load by deflecting routine queries automatically.

Domain Data Sources

- **docs.cisco.com** Cisco Support Documentation → Product Support → select any product → download PDF guides
- **help.ubuntu.com** Ubuntu Help & Manuals → downloadable server/desktop guides
- **docs.aws.amazon.com** AWS Troubleshooting Guides → General Reference → Troubleshooting section
- **support.microsoft.com** Microsoft Support Articles → search any product → export/print to PDF

Sample Instruction Pairs

instruction	response
My router keeps dropping connection every 30 minutes. How do I fix this?	1. Check firmware — update if outdated. 2. Change wireless channel to 6 or 11. 3. Disable QoS under Advanced Settings. 4. Factory reset as last resort.
Ubuntu 22.04 fails to mount external NTFS drive. Error: wrong fs type.	1. Install ntfs-3g: <code>sudo apt install ntfs-3g</code>. 2. Run: <code>sudo ntfsfix /dev/sdX1</code>. 3. Mount: <code>sudo mount -t ntfs-3g /dev/sdX1 /mnt/drive</code>. 4. Check dmesg for hardware errors.
AWS EC2 instance unreachable after security group change.	1. Verify inbound rules allow SSH (port 22) from your IP. 2. Check NACL — both inbound and outbound. 3. Confirm instance state is 'running'. 4. Use EC2 Instance Connect if SSH fails.

Variant 3 — Regulatory Compliance Q&A Assistant

Banking / FinTech / Insurance — Regulatory Intelligence

A bank or NBFC wants compliance analysts to quickly locate and interpret regulatory requirements across hundreds of circulars, guidelines, and frameworks. An LLM fine-tuned on regulatory documents can answer queries such as 'What are RBI's KYC requirements for digital customers?' in seconds — significantly reducing manual lookup time.

Domain Data Sources

- **rbi.org.in** RBI Circulars & Master Directions → Notifications → Master Directions (downloadable PDFs)
- **sebi.gov.in** SEBI Regulations → Legal Framework → Regulations
- **irdai.gov.in** IRDAI Guidelines → Regulations → Guidelines & Circulars
- **bis.org** Basel III Framework → full Basel III text as PDF
- **fatf-gafi.org** FATF Recommendations → Publications → FATF Recommendations PDF

Sample Instruction Pairs

instruction	response
What is the maximum LTV ratio for gold loans as per RBI guidelines?	Per RBI Master Direction on Loans and Advances (2023), LTV for gold loans must not exceed 75% of the value of gold ornaments accepted as collateral.
What are the KYC requirements for a small savings account under RBI norms?	Under RBI Master Direction on KYC (updated 2023), a small account requires a self-certified photograph and one officially valid document. Maximum balance: ₹50,000.
What does SEBI LODR Regulation 17 require for board composition?	SEBI LODR Regulation 17 mandates at least 50% independent directors on the board of listed entities. If the chairperson is non-executive, at least one-third must be independent.

Variant 4 — Clinical Protocol Lookup Assistant

Hospital Systems / Pharma / MedTech — Clinical Decision Support

A hospital network wants to help clinicians, nurses, and pharmacists quickly look up treatment protocols, drug dosage guidelines, and clinical pathways without leaving their workflow. The LLM is fine-tuned to answer protocol-specific questions accurately from clinical literature.

Disclaimer requirement: every generated response must include 'This output is for educational/reference purposes only and must not replace professional clinical judgment.' Add this disclaimer to all instruction-response pairs in your dataset.

Domain Data Sources

- **who.int/publications** WHO Clinical Guidelines → filter by 'guidelines' → download PDFs
- **cdc.gov/mmwr** CDC Clinical Recommendations → downloadable clinical PDFs
- **nice.org.uk/guidance** NICE Guidelines (UK) → clinical guidelines freely downloadable
- **icmr.gov.in** ICMR Guidelines → Guidelines & Publications
- **pubchem.ncbi.nlm.nih.gov** PubChem Drug Data → compound data and drug monographs

Sample Instruction Pairs

instruction	response
What is the first-line treatment for uncomplicated community-acquired pneumonia in adults?	Per WHO guidelines, first-line treatment for uncomplicated CAP is Amoxicillin 500 mg orally 3×/day for 5 days. For penicillin-allergic patients: Doxycycline 100 mg 2×/day. [Educational reference only]
What is the recommended dosing interval for Vancomycin in renal impairment?	For CrCl < 30 mL/min, extend Vancomycin interval to every 24–48 hours with TDM monitoring. Target trough: 10–20 mcg/mL. Consult clinical pharmacist. [Educational reference only]
What are the ICMR criteria for initiating antiretroviral therapy in adults with HIV?	Per ICMR guidelines, ART is initiated when CD4 count < 350 cells/mm ³ or at any CD4 count with WHO Stage 3/4 disease, TB co-infection, or HBV co-infection. [Educational reference only]

Variant 5 — IT Helpdesk & Incident Resolution Bot

Enterprise IT / ITSM / DevOps — Internal Support Automation

A large enterprise IT team manages thousands of endpoints and applications, generating frequent Tier-1 and Tier-2 helpdesk tickets. The ITSM team wants an internal LLM fine-tuned on ITIL documentation, SOPs, and runbooks — generating structured resolution procedures aligned with company policy.

Domain Data Sources

- **axelos.com** ITIL 4 Foundation Guides → ITIL publications (free overview PDFs and sample chapters)
- **learn.microsoft.com** Microsoft Endpoint Manager Docs → Endpoint Manager / Intune documentation
- **docs.servicenow.com** ServiceNow Public Documentation → IT Operations guides
- **access.redhat.com/documentation** RedHat RHEL System Admin Guide → System Administration Guide PDFs
- **ubuntu.com/server/docs** Ubuntu Server Documentation → complete server administration guide

Sample Instruction Pairs

instruction	response
User cannot connect to corporate VPN. Error: 'Authentication failure' on Cisco AnyConnect.	1. Verify AD credentials and MFA token. 2. Flush DNS: <code>ipconfig /flushdns</code>. 3. Reinstall AnyConnect profile if mismatch. 4. Escalate to network team if unresolved after 30 min.
User account locked after 3 failed login attempts. How to unlock in Active Directory?	1. Open ADUC → search user → Properties → Account tab. 2. Uncheck 'Account is locked out'. 3. Reset password if requested. 4. Log incident in ServiceNow with resolution notes.
Disk usage on Linux server at 98%. Which command identifies largest directories?	1. Run: <code>du -sh /* 2>/dev/null sort -rh head -20</code> 2. Check <code>/var/log</code> for oversized logs. 3. Run: <code>journalctl --vacuum-size=500M</code> 4. Raise capacity alert if disk > 90% after cleanup.

Variant 6 — Procurement & Supply Chain Intelligence Assistant

Manufacturing / Retail / Logistics — Enterprise Procurement AI

A manufacturing conglomerate wants to reduce procurement cycle time by enabling buyers and analysts to query procurement policies, supplier specifications, logistics SOPs, and quality standards in natural language. The LLM answers queries like 'What quality checks are mandatory before goods receipt?'

Domain Data Sources

- **iso.org** ISO 9001 Quality Management → ISO 9001 overview + public guidance documents
- **unece.org/trade/uncefact** UN/CEFACT Trade Facilitation → procurement and trade facilitation guides
- **gs1.org/standards** GS1 Supply Chain Standards → global supply chain standards documentation
- **gem.gov.in** India GFR 2017 (Procurement Rules) → General Financial Rules 2017 PDF
- **ascm.org** APICS/ASCM Body of Knowledge → supply chain management overview PDFs

Sample Instruction Pairs

instruction	response
What are the mandatory quality inspection criteria before accepting goods from a new supplier?	Per ISO 9001 Clause 8.4.3: (1) verify against PO specifications, (2) dimensional and quantity inspection, (3) review COA and delivery documents, (4) sampling per AQL table, (5) raise GRN in ERP system.
What is the maximum payment term for MSME suppliers under Indian procurement rules?	Under MSMED Act 2006 and GFR 2017 Rule 144, payment must be made within 45 days of acceptance. Delayed payments attract compound interest at 3× the RBI bank rate.
What documentation is required before raising a purchase order for a new vendor?	Per GFR Rule 144: (1) vendor registration with GST certificate, (2) bank account verification, (3) past performance certificate if > ₹10L, (4) quality assurance plan for technical items.